

Dual-Sided Document Verification & Identity Architecture

Domain: Instant Trust & DPDP Act Compliance • Latency: < 30 Seconds • Verification Cost: Rs. 1.50 – 2.50 / check

1. The Document Verification Trilemma in Indian P2P Platforms

Peer-to-peer property platforms in India encounter three catastrophic failure modes when handling physical spaces: (1) **Renter Fraud**: Students using photoshopped Aadhaar copies or borrowing friends' IDs to vandalize spaces anonymously; (2) **Host / Listing Fraud**: Scammers posting images of spaces they don't own to harvest advance booking fees; and (3) **Data Privacy Violations**: Storing raw unmasked Aadhaar documents on cloud servers violates the **Digital Personal Data Protection (DPDP) Act 2023** and invites massive statutory penalties.

Traditional manual verification takes 24–48 hours and costs Rs. 50–100 per check—destroying instant booking viability. SpaceLoop solves this with an automated, zero-knowledge verification pipeline built entirely on the **India Stack**.

Comparative Architecture: Manual Verification vs. SpaceLoop India Stack

Verification Metric	Traditional Manual Process (Flawed)	SpaceLoop India Stack (Automated)
Processing Latency	24 to 48 hours (manual human review)	Under 30 seconds (Programmatic API execution)
Cost Per Verification	Rs. 50 – 100 per check (staff overhead)	Rs. 1.50 – 2.50 (Direct sandbox API micro-costs)
Forgery Resistance	Low (Easy to photoshop PDF / JPG copies)	100% Cryptographic (Direct UIDAI / DigiLocker fetch)
DPDP Compliance	Critical Risk (Raw Aadhaar images on DB)	100% Compliant (Zero raw Aadhaar stored; masked tokens)
Impersonation Defense	None (Anyone can upload another's ID)	AI Biometric Face Match + Live micro-challenge
Host Ownership Proof	50-page physical deed or zero check	State Discom Electricity API + UPI Penny-Drop

2. Student / Renter Verification Pipeline (Completed in 20 Seconds)

Students undergo a single one-time verification during account creation, executing across three frictionless stages:

Stage 1: Government Identity via DigiLocker / Aadhaar OTP

SpaceLoop connects to DigiLocker and UIDAI sandbox gateways (via certified providers e.g. Setu, Cashfree, Decentro). The student triggers an official UIDAI OTP sent directly to their registered mobile phone. Upon validation, the server receives verified payload data: Legal Name, Date of Birth (verifying 18+ requirement), and Registered State/District. In strict compliance with the DPDP Act 2023, SpaceLoop **never stores raw Aadhaar numbers or image scans**; only a cryptographic SHA-256 hash and masked string ('XXXX-XXXX-4821') are retained.

Stage 2: Academic Affiliation Handshake (Dual Route)

- **Route A (Direct Domain OTP)**: Students from universities with institutional domains enter their email ('@iitd.ac.in', '@du.ac.in', '@vitstudent.ac.in'). SpaceLoop dispatches a 6-digit OTP, validating student status in 5 seconds.
- **Route B (AI OCR for Affiliated Colleges)**: For colleges without custom domains, students snap their physical ID card. Gemini 1.5 Flash performs OCR and cross-references the student name against the DigiLocker payload while checking institutional watermarks, valid academic year, and layout tampering.

Stage 3: AI Liveness & Biometric Face Match (Anti-Impersonation)

To prevent banned students from using a friend's or relative's credentials, the app prompts a 2-second live selfie session. A random dynamic challenge (e.g. 'Blink twice' or 'Tilt head left') neutralizes printed photos and screen replays. An embedded lightweight facial recognition model computes geometric embeddings between the live selfie and the official DigiLocker photo. A confidence threshold of >92% unlocks the **Verified Student Badge**.

3. Property Owner & Host Verification Pipeline (Zero-Scam Guarantee)

To guarantee that renters are not arriving at a non-existent space, unauthorized sublet, or commercial trap, hosts must prove both legal identity and rightful operational control of the physical premises:

Stage 1: State Electricity Board (Discom) Consumer Number API Match

Every legitimate residential or commercial property in India has an active power meter managed by a state or private Discom (e.g., BESCOM Bangalore, TPDDL Delhi, MSEDCL Maharashtra, UPPCL Uttar Pradesh, Adani Mumbai). SpaceLoop integrates with national Discom billing APIs via the Bharat Bill Payment System (BBPS) and state utility gateways:

- Host enters the 10-to-12 digit **Consumer Account (CA) Number** from their latest power bill.
- The API instantly returns the officially registered Consumer Name and Installation Premise Address.
- SpaceLoop algorithmically cross-matches the Discom address with the listing geolocation (<50m tolerance) and matches the registered name against the host's legal ID. This proves physical possession without requiring property deeds.

Stage 2: NPCI UPI Rs. 1 Penny-Drop Bank Account Verification

To prevent anonymous shell accounts from receiving booking revenues, SpaceLoop executes an automated Rs. 1 Penny-Drop via NPCI IMPS/UPI rails to the host's payout account. The banking response returns the official **Beneficiary Name** registered with the bank. If this name matches the host's PAN card and DigiLocker identity, host verification is certified.

Owner Security Check Flow: PAN / Aadhaar Verified • Discom Power Bill CA Match • UPI Bank Penny-Drop Match • Geo-tagged Room Baseline Photos. Result: 100% elimination of ghost listings and phantom hosts.

4. Statutory Data Protection & Legal Enforceability

SpaceLoop is engineered from the ground up to comply with Indian data regulations and legal precedent:

- **Zero Raw Identity Storage:** In full compliance with UIDAI circulars and the DPDP Act 2023, SpaceLoop never writes unmasked 12-digit Aadhaar numbers or raw Aadhaar PDFs to any persistent database. Identities are tokenized into cryptographic hashes.
- **Clickwrap Digital Micro-License:** Each booking programmatically binds both parties to a legally valid, time-limited Revocable Micro-Lease under the Indian Easements Act, 1882. It grants permissive entry only between exact QR check-in and check-out timestamps, preventing any claim of tenancy or squatting rights.
- **Institutional Accountability Deterrence:** Because accounts are tied to verified university credentials and government IDs, students understand that malicious damage or unruly behavior creates immediate legal liability and academic reporting.

5. Micro-Cost Verification Breakdown (Per User Onboarding)

Verification Step	Underlying Technology / Gateway	API Cost	Latency
Aadhaar / DigiLocker OTP	UIDAI / DigiLocker Sandbox (Setu/Decentro)	Rs. 0.80	4.2 sec
Academic Domain / ID OCR	Direct SMTP OTP + Gemini 1.5 Flash OCR	Rs. 0.05	1.8 sec
AI Liveness & Face Match	Lightweight facial embedding matcher	Rs. 0.15	2.1 sec
Discom Electricity Match	BBPS / State Utility Gateway API	Rs. 0.50	3.5 sec
UPI Bank Penny Drop	NPCI IMPS / UPI Penny Drop Gateway	Rs. 1.00	2.8 sec
Total Student Onboarding	Stages 1, 2, and 3 combined	Rs. 1.00	8.1 sec
Total Host Onboarding	Aadhaar + Discom + UPI Penny Drop	Rs. 2.30	10.5 sec